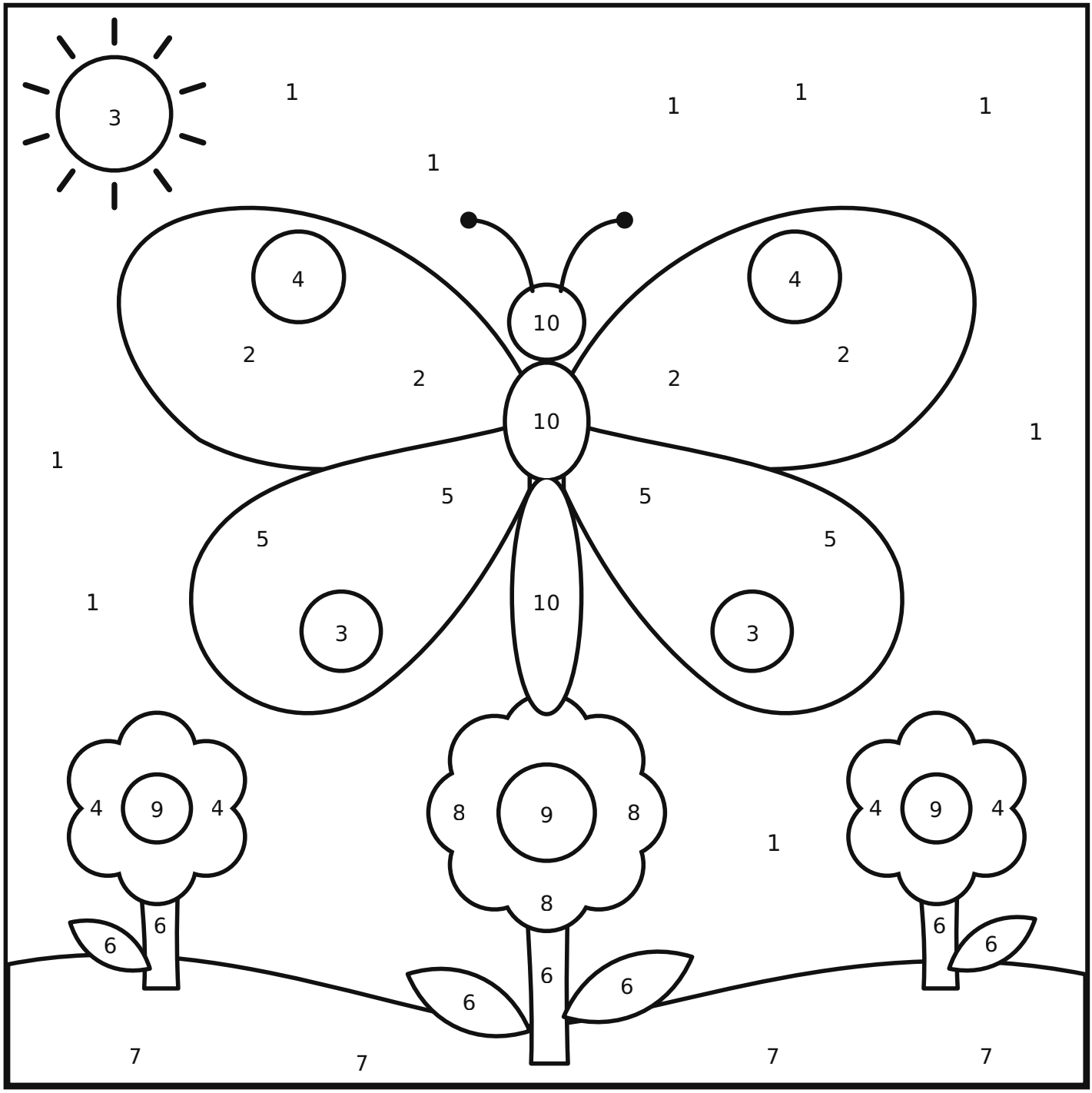
Name: _____ Date: _____

Exponents & Exponential Functions: Color By Number



Color Code:

_____ Green $3x^5$	_____ Brown $x = 4$	_____ Light Green $\frac{1}{x^3}$	_____ Orange $12x^5$	_____ Light Blue x^7
_____ Red 5	_____ Pink $4x^6$	_____ Purple x^5	_____ Black $y = 500(1.2)^x$	_____ Yellow x^{12}

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Directions: Solve each problem and show your work in the box. Your answer should match one of the answers in the Color Code on the coloring sheet. Find the matching answer, write the problem number on its blank, then write the color on the line. Color every part of the picture with that number. Watch out: the answers look alike!

1.
Simplify: $x^4 \cdot x^3$

1. Color: _____

2.
Simplify: $\frac{x^8}{x^3}$

2. Color: _____

3.
Simplify: $(x^3)^4$

3. Color: _____

4.
Simplify: $(2x^3)^2$

4. Color: _____

5.
Simplify: $3x^2 \cdot 4x^3$

5. Color: _____

6.
Simplify: $\frac{18x^7}{6x^2}$

6. Color: _____

7.
Write with a positive exponent: x^{-3}

7. Color: _____

8.
Simplify: $5x^0$ $(x \neq 0)$

8. Color: _____

9.
Solve for x: $3^x = 81$

9. Color: _____

10.
An account starts at $500 and grows 20% each year. Write the exponential function for the balance y after x years.

10. Color: _____